

Introduction to Programming

CIS-130 30 0

Fall Term 2025-2026 School Year Section 30 0 3.00 Credits 08/25/2025 to 12/12/2025
Modified 08/15/2025

Course Description

Introduces students to the terminology, fundamentals and application of the program development process. Basic programming concepts such as problem analysis, logic organization and design, and program development and testing will be implemented. Topics covered include data types, operators, expressions, program flow control statements, and methods.

Outcomes

The student should be able to demonstrate the following outcomes upon successful completion of this course:

1. Explain fundamental programming concepts and terminology.
2. Demonstrate the ability to use an Integrated Development Environment tool (i.e. Visual Studio) efficiently.
3. Design and construct a professional-looking, user-friendly interface that enhances a software solution.
4. Employ a software development process that includes problem analysis, logic and interface design, testing and debugging.
5. Distinguish between a program's input, processing, and output requirements.
6. Demonstrate proper use of a programming language's syntax, data, memory, and control structures.
7. Identify and resolve different types of programming errors.
8. Discuss program logic and code with others.
9. Create a variety of business-related programs that effectively deliver identified requirements.

Technical Skills

- Explain fundamental programming concepts and terminology.
- Demonstrate the ability to use an Integrated Development Environment tool (i.e. Visual Studio).
- Demonstrate proper use of a programming language's syntax, data, memory, and control structures.

Critical Thinking

- Distinguish between a program's input, processing, and output requirements.
- Identify and resolve logic, syntax, runtime, and arithmetic errors.

Professionalism

- Design and construct a professional-looking, user-friendly interface that enhances a software solution.
- Employ a software development process that includes problem analysis, logic and interface design, testing and debugging.

Communication

- Demonstrate the ability to discuss program logic and code with peers.

Additional Outcomes

Course Materials

Starting Out with Visual C#

- **Author:** Tony Gaddis
- **Publisher:** Pearson
- **Edition:** 6th
- **ISBN:** 978-0-13-808756-2
- **Availability:** Campus Bookstore; Amazon

Deliverables

Final Project

Students will demonstrate the ability to create a Windows Forms application that utilizes several key aspects of programming, such as variables, loops, conditionals, methods, arrays, and classes.

Evaluation Procedures and Grading

Type	Weight
Tutorials	13.5%
Program Problems	34.5%
Participation	5.5%

Weekly Quiz	10.3%
Exam/Project	34.5%
Milestones	1.7%
Total	100.0%

Grade	Range	Notes
A	90 to 100	
B	80 to 89.99	
C	70 to 79.99	
D	60 to 69.99	
F	0. to 59.99	

Resulting grade and related performance levels

Additional Items

Course Outline

Week	Read	Tutorials	Program Problem	Quiz	Participation	Milestone	Exam / Project
1	Chapter 1	yes		yes	yes		
2	Chapter 2	yes	yes	yes	yes		
3	Chapter 3	yes	yes	yes	yes		
4	Chapter 3	yes	yes	yes	yes		
5	Chapter 4	yes			yes		yes

6	Chapter 4	yes	yes	yes	yes		
7	Chapter 5	yes	yes	yes	yes		
8	Chapter 5	yes	yes	yes	yes		
9	Chapter 6	yes			yes		yes
10	Chapter 6	yes	yes	yes	yes		
11	Chapter 7	yes	yes	yes	yes		
12	Chapter 7	yes	yes	yes	yes	yes	
13					yes	yes	yes
14	Chapter 8		yes	yes	yes	yes	
15	Chapter 10	yes		yes	yes	yes	
16	Chapter 12						yes

When	Topic	Notes
Unit 1		Chapter 1: Intro to Computers and Programming Chapter 2: Intro to Visual C# Chapter 3: Processing Data
Unit 2		Chapter 4: Making Decisions Chapter 5: Loops, Files, Random Numbers
Unit 3		Chapter 6: Methods/Modularity Chapter 7: Arrays/Lists
Unit 4		Chapter 8: Text Processing Chapter 10: Intro to Classes